



Protege Knowledge Base and Reasoning

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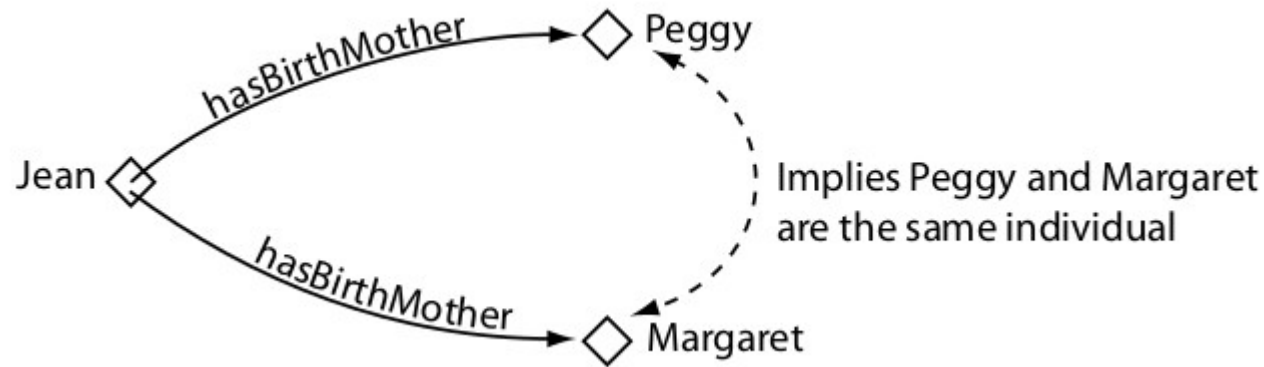


Protégé

- <http://protege.stanford.edu/>
- Protégé supports **OWL 2** and RDF.
- Protégé is based on **Java**.
- It technically helps to build a knowledge base but conceptually is on authors's side.

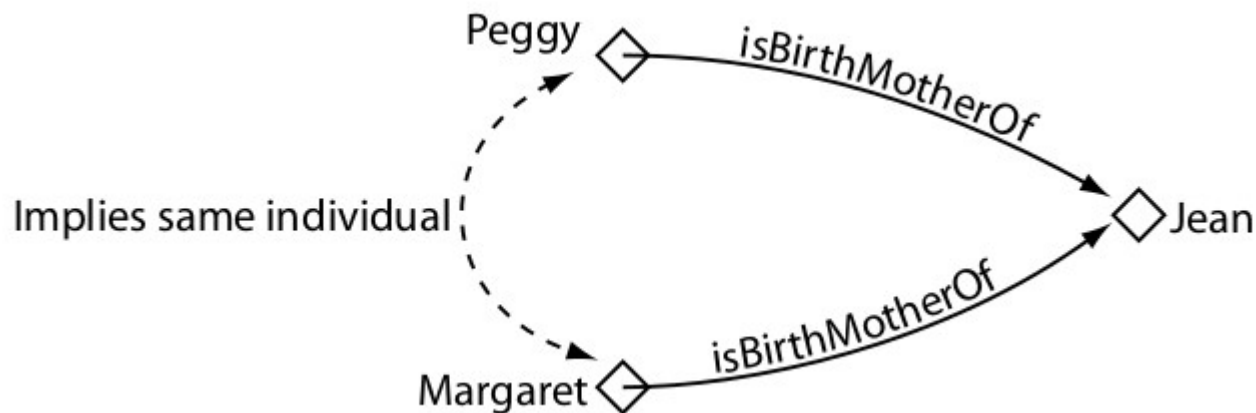


- **Functional properties** are also known as single valued properties and also features.



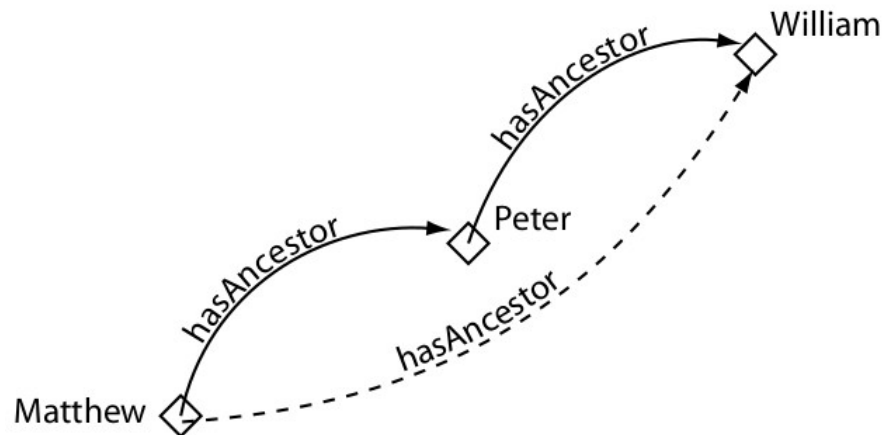


- **Inverse Functional Properties**
- If a property **is inverse functional** then it means that the inverse property **is functional**
-



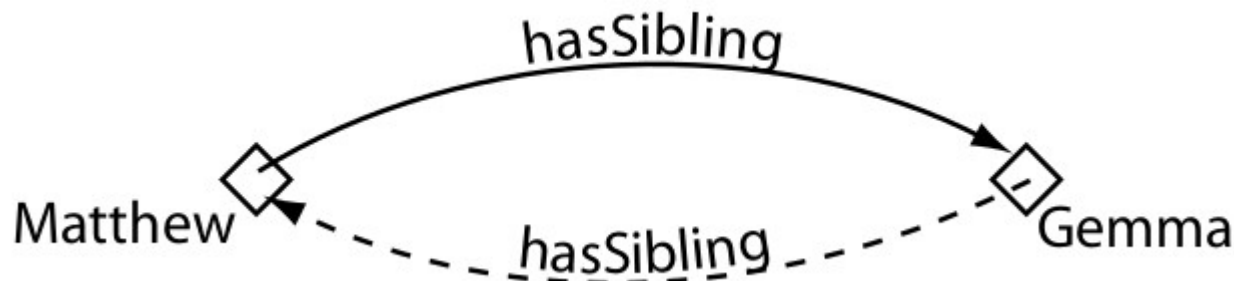


- **Transitive Properties**
- If a property is transitive, and the property relates individual a to individual b, and also individual b to individual c, then **we can infer that individual a is related to individual c via property P**



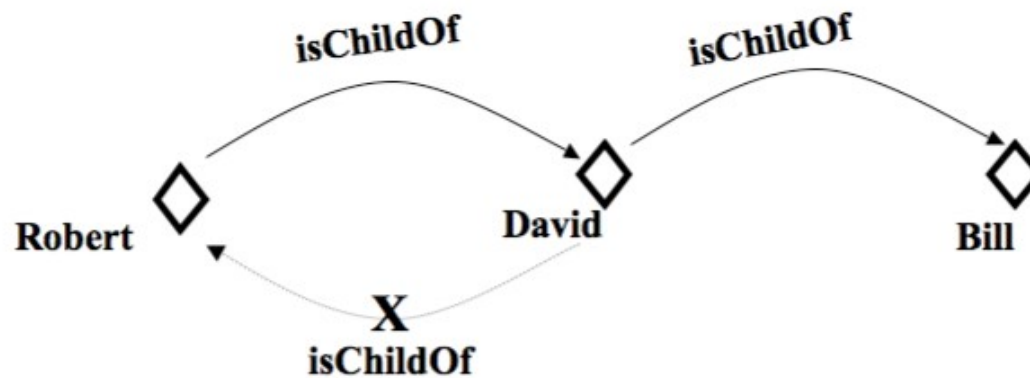


- **Symmetric Properties**
- If a property P is symmetric, and the property relates individual a to individual b then individual b is also related to individual a via property P



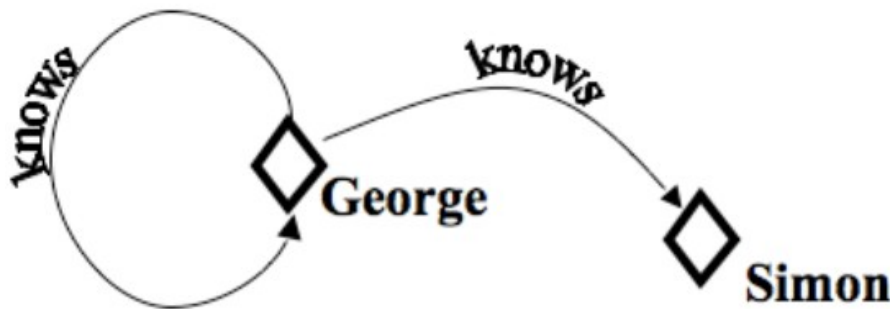


- **Asymmetric properties**
- If a property P is antisymmetric, and the property relates individual a to individual b then individual b **cannot be related to individual a via property P**



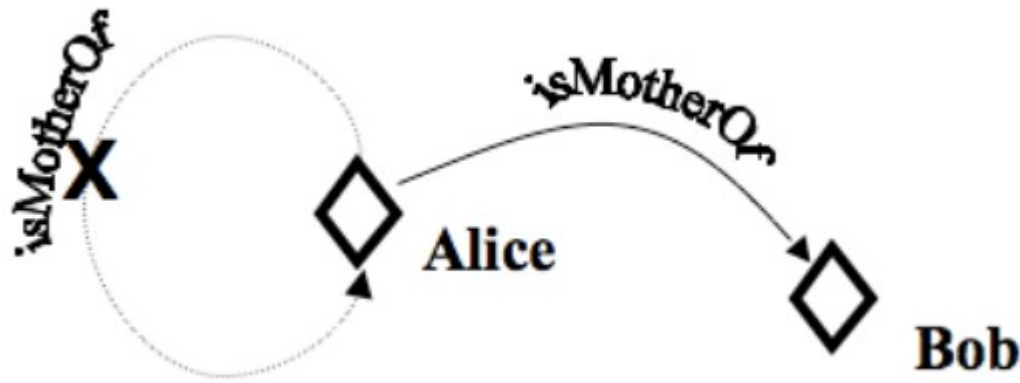


- Reflexive properties
- A property P is said to be reflexive when the property must relate individual a to itself





- Ir-reflexive properties
- If a property P is irreflexive, it can be described as a property that relates an individual a to individual b, where individual a and individual b are not the same





OWL Constructor	DL Syntax	Manchester OWL S.	Example
intersectionOf	$C \sqcap D$	C AND D	Human AND Male
unionOf	$C \sqcup D$	C OR D	Man OR Woman
complementOf	$\neg C$	NOT C	NOT Male
oneOf	$\{a\} \sqcup \{b\} \dots$	$\{a \ b \ \dots\}$	{England Italy Spain}
someValuesFrom	$\exists R \ C$	R SOME C	hasColleague SOME Professor
allValuesFrom	$\forall R \ C$	R ONLY C	hasColleague ONLY Professor
minCardinality	$\geq N \ R$	R MIN 3	hasColleague MIN 3
maxCardinality	$\leq N \ R$	R MAX 3	hasColleague MAX 3
cardinality	$= N \ R$	R EXACTLY 3	hasColleague EXACTLY 3
hasValue	$\exists R \ \{a\}$	R VALUE a	hasColleague VALUE Matthew



- Look up some examples from previous slides

Restricted to FATISDA



Simple Step*

- Determine Class and its attribute
- Determine individual of each class
- Determine object property
- Determine data property
- Check

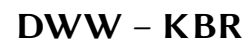


- Astronomi Ontology
 - Class: Person, Astronomer, Astronomical_Model
 - Object Property: invent, agree

Restricted to FATISD



- Netherlands (N) colonized (c) Indonesia(I) from 1595 until 1942.
- Next, Japan(J) colonized Indonesia from 1942 until 1945.
- Indonesia's independence day is August 17, 1945. It was Friday. Soekarno (S) and Muhammad Hatt(MH)a were the declarators of the independence. The location was at Soekarno's house (SH).The address was at Jl.Pegangsaan Timur no.56, Jakarta. The first location plan was at Lapangan Banteng(LB). Soekarno was sick on that day.





- Class: Country, Person, Location, Day, HealthCondition
- ObjectProperty: hasColonized, hasProclamator, hasLocation, hasPlannedLocation, hasCondition
- DataProperty: from, end, hasAddress, onDay, nameDay



Further Reading

- A Semantic Web Primer. The MIT Press Cambridge, Massachusetts, London, England. Grigoris Antoniou and Frank van Harmelen.
- <http://www.w3.org/TR/owl2-primer/>
- Ontology Development 101: A Guide to Creating Your First Ontology Natalya F. Noy and Deborah L. McGuinness Stanford University, Stanford, CA, 94305
- Practical Guide of Protege (many open sources)
- Manch Syntax